

COMPUTATION AND VERIFICATION REPORT

QIAN QI
Peking University

1. FIXED INPUTS AND CHRONOLOGY

The report answered is the R44 external report dated September 25, 2026, at review commit `bdfd4e94b74a41ef904158d0061d10a6fcf39c8c`. The development baseline is R46 commit `fb88b1122ce5709d36104993937083b57470d1de`. The new manuscript is R47, dated September 26, 2026. These identities separate the report, the latest numerical development, and the new publication. The remote branch is `revision/econometrica-r47-complete-referee-2026-09-26`.

The original R44 protocol and all sixteen primitive model files are unchanged. The R46 protocol explicitly describes a retrospective price experiment; its extension records that the full search was developed after price and root outputs had been observed. Neither is relabeled a fresh holdout. The original R45 draft and its complete historical content are retained as ordinary authoring source and a separate reading volume. The inherited numerical code and proof objects remain available at their original repository paths.

2. REPLAYED AND NEWLY CONSTRUCTED OBJECTS

There are sixty-four inherited R46 objects: thirty-two constant/restart-price certificates, sixteen price-strengthened roots, and sixteen complete price-strengthened trees. The present revision replays every object using the independently implemented readers. The current local replay accepted all sixty-four and recorded per-object hashes and elapsed time. It did not rerun the optimizers or change any frozen endpoint.

The new witness constructor creates sixty-four further objects, four precision levels on each unchanged model. These are not sixty-four independent environments. Each includes the operating lower and upper witnesses, nonnegative prices inherited from a checked source certificate, a directed lower potential, one common repaired policy, its implementation cost, and explicit error budgets. A separate standard-library reader checks all of these without importing the constructor or a solver. It uses an unshifted potential representation to reconstruct the lower recursion. It also checks operating sub-/supersolution inequalities, probability simplices and grids, original model/protocol hashes, and the stronger upper-witness all-restart inequalities. All sixty-four pass.

Fifty-one deliberately malformed witness objects are rejected, with three valid controls accepted. The mutations separately alter endpoints, probabilities, policy length, terminal/operating enclosures, transformed potentials, cost budgets, model/source/protocol identity, target, and precision. They are smoke tests, not a proof of verifier completeness. The inherited price/tree property and mutation tests remain distinct. The mathematical proofs establish the general results; finite replay establishes only the declared finite objects and arithmetic contracts.

3. SUCCESS CRITERIA AND EXCLUSIONS

The primary criterion is exact absolute width at most $1/1000$. Original frozen B and A counts are four and five of twelve primary cases, with no tie successes. The retrospective price tree attains six primary and three tie targets. The certified scalar portfolio attains nine of sixteen. In

the witness-only precision audit four targets are met at each of four precisions; those intervals use the price-only lower, not the stronger tree lower. IO is zero cost; T0's operating value is exactly representable. Neither is described as a difficult approximate-oracle test.

SCIP native lower bounds are excluded from every exact portfolio. Only its independently evaluated feasible policy may supply an upper endpoint. A newly reconstructed price certificate is not a post-certification of that solver's nonlinear global lower value. No native bound has been promoted by a name change. The original thirty positive-cost continuous intervals and the nonlinear experiments are not newly closed, and no empirical calibration is asserted.

4. REPRODUCTION COMMANDS AND DATA BOUNDARY

From the repository root, the complete publication can be rebuilt with

```
bash R47_BUILD.sh
```

The build does not rerun the original optimizers. Its central commands are

```
python revisions/2026-09-26-r47/replication/replay_r46.py
python revisions/2026-09-26-r47/replication/witness_audit.py
python revisions/2026-09-26-r47/replication/verify_witness.py
python revisions/2026-09-26-r47/replication/test_witness.py
python revisions/2026-09-26-r47/replication/tabulate_r47.py
```

The price replay needs the inherited R44 and R46 readers and models. The witness construction and reader use only Python standard-library exact arithmetic. Figure regeneration additionally uses matplotlib. PDF production uses the retained, unmodified Econometrica `econsocart` class and `pdfLaTeX/latexmk`. Python package versions, the TeX build output, platform metadata, and all final content hashes are recorded by the publication build.

The source transfer is verified before extraction into normal files. No transport chunk is an article, and no restoration operation is needed by a referee to read the final PDFs or source tree. The publication manifest records source, table, proof, result, and PDF hashes. The inherited-file check verifies that the baseline tracked objects are not modified or removed by publication. The final Git commit gives the immutable review identity; a manifest does not try to contain its own hash or its future commit hash.

5. THEOREM-TO-CODE MAP

Analytic statement	Finite evidence and remaining boundary
Restart-price lower and gap	Independent price replay; exact rational gap decomposition for sixteen saved policies. Borel integration/measurability is proved analytically, not machine checked.
Price cuts and complete covers	Independent price-tree reader rebuilds cuts and verifies every covering split and leaf. Eventual completeness is not a measured complexity bound.
Witness lower error	Independent operating sub-/supersolutions and transformed-potential checks on sixty-four precision objects; exact lower-loss budget checked.
Upper-witness repair	One policy evaluated against the upper witness at every restart; sufficient-slack budget checked where applicable. Three coarse-precision failures of that sufficient guard retain actual valid row certificates.
Controlled-state density	Mathematical feasible-upper approximation; no executed paired continuum closure is inferred.
Structural aggregation/fibers	Preserved results under their own assumptions; not replacements for generic controlled-state convergence.
SCIP native lower	Uncertified numerical diagnostic, outside the exact-reader boundary.